HAIER EGYPT

Sales Analytics Project Documentation

End-to-end workflow: web-scraped reference data, synthetic transaction generation, data quality checks, derived fields, and dashboard reporting

Project scope

The project uses scraped public product information as a reference point to build a realistic synthetic order dataset for analytical practice. The transaction records are simulated and should not be presented as official Haier Egypt sales data.

Executive Summary

This project demonstrates a complete data analytics workflow, starting with web scraping to establish realistic product and pricing context, followed by synthetic data generation to create the order-level history needed for business analysis. The generated dataset was structured to represent common sales operations, including product categories, sales partners, channels, discounts, delivery outcomes.

The dataset was then prepared for analysis through data profiling, cleaning rules, validation checks, and date-based helper columns. An Excel dashboard was created to summarize overall performance, sales trends, regional contribution, partner performance, and product/category results.

Project at a Glance

Measure	Current workbook snapshot
Order records	19,847
Reporting period	1 January 2025 – 15 October 2026
Product IDs / models	59
Sales partner IDs	27
Regions / city labels	5 regions / 23 cities
Sales channels	4
Dashboard tabs in the uploaded workbook	Overview, Sales Trend, Products

Dashboard KPI Snapshot

KPI	Value shown in workbook
Total revenue	EGP 739,310,000
Profit reported by current model	EGP 291,505,001
Total discounts	EGP 47,641,199
Total cost reported by current model	EGP 400,163,800
Units sold	27,069
Profit margin displayed	39.4%

The revenue and discount totals reconcile to net sales of EGP 691,668,801. Profit and margin are reproduced here as displayed in the workbook; the cost/profit calculation requires the final validation described in the QA section.

1. End-to-End Project Workflow

1. **Reference data collection.** Public product information was gathered through web scraping to establish real-world context for product names, model codes, categories, and plausible price ranges. The scraped reference data was used as a foundation, not as a source of actual order transactions.
2. **Data profiling and scoping.** The collected fields were reviewed to understand their structure and limitations. Because product pages do not provide a full sales ledger with order dates, customers, partners, discounts, delivery outcomes, and campaign history, additional transaction-level data was required for dashboard analysis.
3. **Synthetic data generation.** A larger order-level dataset was generated using the scraped product context as a reference. It introduced realistic transaction fields such as Order_ID, Order_Date, Region, City, Sales_Channel, Sales_Partner, Customer_Segment, Product_ID, Units_Ordered, prices, discounts, cost, payment method, order status, return flag, delivery days, and promotion name.
4. **Dirty-data scenario.** The generated data included imperfections to simulate a practical analytics assignment: placeholder values, inconsistent text formatting, unusual quantity values, and fields that required business-rule validation. This created a reason to assess and document data quality before reporting.
5. **Cleaning and transformation.** The dataset was organized for analysis by reviewing field types, identifying placeholder values, standardizing labels, checking duplicate identifiers, validating calculations, and deriving date attributes for period-based analysis. Uncertain details should be flagged rather than fabricated.
6. **Dashboard development.** The prepared data was loaded into an Excel-based reporting model using PivotTables and charts to create three visible dashboard pages: Overview, Sales Trend, and Products. Supporting data and pivot sheets are hidden in the uploaded workbook.

2. Dataset Structure

The main data table contains 29 columns. They cover identifiers, date/location information, commercial dimensions, customer and product descriptors, financial fields, operational status, and date helpers.

Field group	Representative columns	Analytical purpose
Identifiers	Order_ID, Customer_ID, Product_ID, Sales_Partner_ID	Keep orders and entities distinguishable; support counting and grouping.
Time and geography	Order_Date, Region, City	Trend analysis and regional comparisons.
Commercial dimensions	Sales_Channel, Sales_Partner_Name, Customer_Segment	Compare channel, partner, and customer mix.
Product attributes	Category, Model_Code, Units_Ordered	Understand category, model, and volume performance.
Financial fields	Unit_Price_EGP, Order_Amt, Discount_Pct, Discount_Amt, Unit_Cost_EGP, Profit_Per_Order	Measure revenue, discounting, costs, and profitability.
Operational fields	Payment_Method, Order_Status, Return_Flag, Delivery_Days, Promotion_Name	Review order outcomes, fulfilment, returns, payments, and promotions.
Date helpers	Order_Date (Year), (Quarter), (Month Index), (Month)	Support chronological aggregation and correctly sorted month visuals.

Why synthetic data was necessary

The scraped product reference data supplied realism around the product catalogue, but it could not by itself answer questions about monthly sales, delivery status, discount totals, customer segments, or partner contribution. Synthetic transaction records made those analytical dimensions available while keeping the distinction between reference data and simulated sales explicit.

3. Data Cleaning and Validation Workflow

The cleaning approach focused on making each field interpretable, keeping business logic explicit, and preventing formatting problems from creating misleading categories or KPIs.

Step	Cleaning / validation rule	Why it matters
1. Profile the data	Review row count, field names, data types, unique values, ranges, and placeholder frequency.	Establish a baseline before changing values.
2. Set data types	Parse Order_Date as a date; keep IDs as text; use numeric types for units, prices, discounts, and delivery days.	Avoid incorrect sorting, arithmetic, or aggregation.
3. Standardize labels	Use consistent spelling and capitalization for categories, cities, regions, partners, channels, payment methods, and statuses.	Avoid splitting one business category into multiple labels.
4. Handle unknowns	Treat explicit placeholders such as “N/A” as unknown values; do not invent customer, partner, or location details.	Protect data integrity while making missingness visible.
5. Check duplicates	Check Order_ID uniqueness and determine whether repeated rows represent duplicates or legitimate transactions.	Prevent inflated order and revenue totals.
6. Apply status-aware rules	Interpret zero revenue on Cancelled/Returned orders separately from missing or erroneous revenue on active orders.	Avoid treating every zero as the same type of data issue.
7. Validate calculations	Reconcile order amount, discount amount, net order amount, total cost, and profit to explicit formulas.	Keep derived KPIs consistent and auditable.
8. Review outliers	Flag unusually high Units_Ordered values for business review rather than removing them automatically.	Distinguish data-entry errors from legitimate bulk orders.

4. Custom and Calculated Columns

Column	Meaning / intended logic
Order_Amt	Gross order value. Generally Units_Ordered × Unit_Price_EGP for sales-bearing orders; status rules may set cancelled or returned orders to zero.
Discount_Amt	Discount amount: Order_Amt × Discount_Pct.
Order_After_Discount	Net order value after discount: Order_Amt − Discount_Amt, equivalent to Order_Amt × (1 − Discount_Pct).
Profit_Per_Order	Intended profit: Order_After_Discount − total order cost. If Unit_Cost_EGP is a true per-unit cost, total order cost should be Unit_Cost_EGP × Units_Ordered.

Calculation principle: Keep gross revenue, discounts, net revenue, total cost, and profit separate. A field named Unit_Cost_EGP should normally be multiplied by Units_Ordered to obtain total cost per order. The current workbook’s cost/profit aggregation needs a final reconciliation before profit-based business conclusions are used.

5. Dashboard Page 1 — Overview

Purpose: provide a quick executive snapshot of the overall simulated business and give the reader a starting point before moving into trends or category-level detail.

Key performance indicators

- Total Sales / Revenue — gross order value in EGP.
- Total Orders — number of order records in the model.
- Total Profit and Profit Margin — the profitability values currently calculated in the workbook; interpret after cost validation.
- Units Sold — total quantity across the order records.
- Total Discount and Total Cost — summarize discounting and cost values in the current model.
- Number of Sales Partners — number of partner IDs represented in the dataset.

Visuals and how they help

Visual	Question answered	Business use
Order Status	How many orders are Delivered, Delayed, Cancelled, or Returned?	Monitor fulfilment performance and identify where operational review may be needed.
Monthly Sales & Profit Trend	How do revenue and reported profit vary by month?	Identify trend patterns and periods that may need deeper investigation.

The uploaded dataset contains 17,644 Delivered orders out of 19,847 records (approximately 88.9%). This provides a useful high-level status indicator, while delayed, cancelled, and returned records should be inspected separately for operational interpretation.

6. Dashboard Page 2 — Sales Trend

Purpose: investigate sales performance over time, compare geographic contribution, and identify the partners contributing most to revenue.

Visual	What it shows	How it supports decisions
Monthly Profit Trend	Reported profit by month.	Shows periods that warrant a closer look; use only after validating the profit formula.
Quarterly Performance	Net revenue and reported profit by quarter.	Makes quarter-to-quarter comparison easier and highlights potential seasonality.
Top 10 Sales Partners by Revenue	The ten partner names with the highest revenue in the current model.	Supports partner performance reviews, account planning, and prioritization.
Revenue by Region	Revenue contribution across five regions.	Helps compare regional demand and inform local campaigns or inventory planning.

Useful reading from the current data

- Greater Cairo has the largest reported regional revenue at approximately EGP 242.9M, followed by Delta at EGP 187.8M.
- The Top 10 partner visual allows the team to compare high-contribution partners without losing sight of the overall regional picture.
- The workbook places partner analysis within the Sales Trend page. The uploaded file does not contain a separate Sales Partners worksheet.

7. Dashboard Page 3 — Products

Purpose: compare product categories on revenue, volume, and reported profitability, and inspect the relationship between revenue and net revenue at transaction level.

Visual	What it shows	Why it is useful
Revenue by Category	Gross revenue contributed by each product category.	Identifies the categories generating the greatest commercial value.
Units Sold by Category	Quantity sold in each category.	Adds volume context: a category with high revenue may reflect price, quantity, or both.
Profit Margin by Category	Reported margin by category.	Can highlight differences in profitability, provided cost and profit calculations are validated.
Revenue vs Net Revenue Scatter	Transaction-level relationship between total revenue and net revenue after discounts.	Shows how discounting affects gross-to-net values and highlights unusual observations.

Product/category signals in the current workbook

Category	Reported revenue	Units sold
Air Conditioner	EGP 222.10M	6,542
Refrigerator	EGP 173.54M	5,326
Washing Machine	EGP 130.91M	5,275
TV (label shown as “Tv” in the file)	EGP 82.06M	4,814
Dishwasher	EGP 79.46M	2,710
Freezer	EGP 51.24M	2,402

Air Conditioners have the highest reported revenue and unit volume in the current dataset. These are descriptive findings fro m simulated data, not statements about actual Haier sales. Category-level margin comparisons should be treated as provisional until the cost logic is reconciled.

8. KPI Definitions

KPI	Definition used / recommended
Total Revenue	Sum of Order_Amt. Confirm that the measure follows the chosen status rules.
Net Revenue / Sales After Discount	Sum of Order_After_Discount; should equal gross revenue minus discounts.
Total Discount	Sum of Discount_Amt.
Units Sold	Sum of Units_Ordered, with treatment of cancelled/returned records explicitly defined.
Total Cost	Sum of total order cost. If Unit_Cost_EGP is per unit, calculate Unit_Cost_EGP × Units_Ordered before summing.
Total Profit	Net revenue minus total cost, or sum of order-level profit using that same cost logic.
Profit Margin	Define the denominator explicitly. The uploaded dashboard appears to use reported profit ÷ gross revenue.
Order Count	Count of unique Order_ID values.

9. Project Value and Conclusion

The project brings together the key stages of an applied analytics workflow: acquiring reference information through web scraping, translating it into a realistic synthetic transactional dataset, assessing and cleaning data, engineering useful da te and financial fields, and presenting results through a multi-page Excel dashboard.

The dashboard provides a structured view of overall sales, order outcomes, monthly and quarterly trends, region and partner contribution, and category-level performance. The main deliverable is not only the dashboard itself, but also the documented path from raw reference information to an analysis-ready data model, with clear assumptions and validation rules.